

MailMind

Complete Oracle Cloud Deployment

Guide

Step-by-Step Instructions for Multi-User Production Deployment

Generated: June 09, 2026

Estimated Deployment Time: 1.5 - 2 hours

Table of Contents

SECTION 1: Gather Your API Credentials (30 minutes)

SECTION 2: Oracle Cloud Account Setup (20 minutes)

SECTION 3: Create VM Instance (15 minutes)

SECTION 4: Open Firewall Ports (10 minutes)

SECTION 5: SSH into Your Instance (10 minutes)

SECTION 6: Install Docker (10 minutes)

SECTION 7: Install Docker Compose (5 minutes)

SECTION 8: Clone Repository (5 minutes)

SECTION 9: Create Environment File (10 minutes)

SECTION 10: Create docker-compose.yml (5 minutes)

SECTION 11: Create Dockerfile (5 minutes)

SECTION 12: Create nginx.conf (5 minutes)

SECTION 13: Start Services (10 minutes)

SECTION 14: Get SSL Certificate (15 minutes)

SECTION 15: Point Your Domain (10 minutes)

SECTION 16: Deploy Frontend to Vercel (15 minutes)

SECTION 17: Verify Everything Works (10 minutes)

SECTION 18: Maintenance & Monitoring

SECTION 19: Troubleshooting

SECTION 20: Final Checklist

SECTION 1: Gather Your API Credentials (30 minutes)

Step 1.1: Get Azure Credentials

You need Azure Active Directory credentials for Microsoft Graph access. Follow these steps carefully:

1. Go to **portal.azure.com**
2. Search for **"App registrations"** → Click it
3. Click **"New registration"**

☞ Name: MailMind

☞ Supported account types: Accounts in any organizational directory

☞ Redirect URI: Web → `https://mailmind.yourdomain.com/auth/microsoft/callback`

☞ Click **Register**

4. Save these values:

☞ AZURE_CLIENT_ID = Application (client) ID

☞ AZURE_TENANT_ID = Directory (tenant) ID

5. Click **Certificates & secrets** (left menu)

☞ Click **New client secret**

☞ Description: MailMind deployment

☞ Expires: 24 months

☞ Click **Add**

☞ Copy the Value: AZURE_CLIENT_SECRET = [the long string]

6. Find your Azure OpenAI resource:

☞ Go to portal.azure.com → Search "OpenAI"

☞ Click your OpenAI resource

☞ Left menu → Keys and Endpoint

☞ Save: AZURE_OPENAI_API_KEY and AZURE_OPENAI_BASE_ENDPOINT

[OK] Save all 5 Azure values in a text file

Step 1.2: Get Google Credentials

1. Go to **console.cloud.google.com**

2. Create new project:

☞ Click project dropdown (top left)

â€¢ Click **NEW PROJECT**

â€¢ Name: MailMind

â€¢ Click **CREATE**

3. Enable Gmail API:

â€¢ Search bar â†’ Type "Gmail API" â†’ Click it

â€¢ Click **ENABLE**

4. Create OAuth credentials:

â€¢ Left menu â†’ **Credentials**

â€¢ Click **Create Credentials** â†’ **OAuth client ID**

â€¢ Application type: **Web application**

â€¢ Name: MailMind

â€¢ Authorized redirect URIs:

- <https://mailmind.yourdomain.com/auth/google/callback>

- <http://localhost:3000/auth/google/callback>

â€¢ Click **CREATE**

5. Copy: `GOOGLE_CLIENT_ID` and `GOOGLE_CLIENT_SECRET`

[OK] Save both Google values

Step 1.3: Generate Random Approval Token

On your local machine, run one of these commands to generate a random 32-character hex string:

Mac/Linux:

```
openssl rand -hex 32
```

Windows (PowerShell):

```
[System.Convert]::ToHexString((New-Object  
System.Security.Cryptography.RNGCryptoServiceProvider).GetBytes(16))
```

Copy the output (32-character hex string):
APPROVAL_TOKEN = [your-random-string]
[OK] Save this value

SECTION 2: Oracle Cloud Account Setup (20 minutes)

Step 2.1: Create Oracle Account

1. Go to **oracle.com/cloud/free**
 2. Click **Start for free**
 3. Fill in registration form:
 - Email address
 - Password
 - Cloud account name: mailmind
 - Country
 4. Click **Verify my email** (check your inbox)
 5. Complete email verification
 6. Add payment method (credit card required but won't be charged)
 7. Click **Start my free trial**
- Wait 5-10 minutes for activation. Check your email for confirmation.

Step 2.2: Log into Oracle Cloud Console

1. Go to **cloud.oracle.com**
2. Click **Sign In**
3. Select your region (top right):
 - Choose closest to you (e.g., us-phoenix-1, us-ashburn-1)
4. Enter email and password
5. You're now in the Oracle Cloud Console!

SECTION 3: Create VM Instance (15 minutes)

Step 3.1: Navigate to Compute

1. Click **Hamburger menu** (top left) → **Compute** → **Instances**
2. Click **Create instance**

Step 3.2: Configure Instance

Setting	Value
Name	mailmind-api
Image	Ubuntu 22.04 (free tier eligible)
Shape	VM.Standard.E2.1.Micro or E3.1.Micro
OCPU	1
RAM	1 GB
VCN	Create new VCN (mailmind-vcn)
Subnet	Create new subnet (mailmind-subnet)
Public IP	Assign a public IPv4 address &œ...
SSH Key	Generate a key pair for me

After clicking **CREATE INSTANCE**:

1. Wait for instance state to say **RUNNING** (green)
2. Click on instance name → Instance details
3. Find **Primary public IP address** and copy it (e.g., 152.70.123.45)
4. Save your SSH private key (named ssh-key-YYYY-MM-DD.key)
&œ... Save both the IP address and SSH key file

SECTION 4: Open Firewall Ports (10 minutes)

Step 4.1: Add Security Rules

1. In Oracle Console, left menu: **Networking** → **Virtual cloud networks**
2. Click **mailmind-vcn**
3. Under Resources, click **Security lists**
4. Click **Default Security List for mailmind-vcn**

Step 4.2: Add Three Ingress Rules

Rule Name	Protocol	Port	Source
HTTP	TCP	80	0.0.0.0/0
HTTPS	TCP	443	0.0.0.0/0
SSH	TCP	22	0.0.0.0/0

For each rule:

- Click **Add ingress rules**
- Set Stateless: ☐ (unchecked)
- Source Type: CIDR
- Source CIDR: 0.0.0.0/0
- IP Protocol: TCP
- Destination Port Range: [port number from table]
- Click **Add**

... All three rules added

SECTION 5: SSH into Your Instance (10 minutes)

Step 5.1: Connect via SSH (Mac/Linux/Windows)

Open terminal on your local machine:

```
# Navigate to where you saved the key
cd ~/Downloads # or wherever you saved ssh-key-YYYY-MM-DD.key

# Set correct permissions
chmod 600 ssh-key-YYYY-MM-DD.key

# SSH in (replace IP with your instance IP)
ssh -i ssh-key-YYYY-MM-DD.key ubuntu@152.70.123.45
```

You should see:

```
Welcome to Ubuntu 22.04.3 LTS...
ubuntu@mailmind-api:~$
```

For Windows PowerShell:

```
$key = "C:\Users\YourName\Downloads\ssh-key-YYYY-MM-DD.key"
ssh -i $key ubuntu@152.70.123.45
```

âœ… You're now in the Oracle VM!

SECTION 6: Install Docker (10 minutes)

In the SSH session (you should see ubuntu@mailmind-api:~\$):

```
# Update system packages
sudo apt-get update
sudo apt-get upgrade -y

# Download Docker installer
curl -fsSL https://get.docker.com -o get-docker.sh

# Run installer
sudo sh get-docker.sh

# Add ubuntu user to docker group
sudo usermod -aG docker ubuntu

# Logout and log back in
exit
```

Back on your local machine, SSH back in:

```
ssh -i ssh-key-YYYY-MM-DD.key ubuntu@152.70.123.45
```

Verify Docker works:

```
docker --version
# Should print: Docker version 24.0.0 (or similar)

docker run hello-world
# Should print: Hello from Docker!
```

SECTION 7: Install Docker Compose (5 minutes)

Still in SSH session:

```
# Download Docker Compose
sudo curl -L
"https://github.com/docker/compose/releases/latest/download/docker-compose-$(uname -s)-$(uname
-m)" -o /usr/local/bin/docker-compose

# Make it executable
sudo chmod +x /usr/local/bin/docker-compose

# Verify
docker-compose --version
# Should print: Docker Compose version 2.20.0 (or similar)
```

âœ… Docker and Docker Compose installed

SECTION 8: Clone Repository (5 minutes)

In SSH session:

```
# Clone your GitHub repo
git clone https://github.com/YOUR-USERNAME/mailmind.git

# Enter directory
cd mailmind

# Verify structure
ls -la
# Should see: backend/ frontend/ docker-compose.yml
```

SECTION 9: Create Environment File (10 minutes)

In SSH session, in the mailmind directory:

```
cat > .env << 'EOF'
# Database
DATABASE_URL=postgresql://mailmind:mailmind-secure-pwd-2024@postgres:5432/mailmind
DB_HOST=postgres
DB_USER=mailmind
DB_PASSWORD=mailmind-secure-pwd-2024
DB_NAME=mailmind

# Microsoft Azure
AZURE_CLIENT_ID=YOUR-AZURE-CLIENT-ID
AZURE_TENANT_ID=YOUR-AZURE-TENANT-ID
AZURE_CLIENT_SECRET=YOUR-AZURE-CLIENT-SECRET
AZURE_OPENAI_API_KEY=YOUR-AZURE-OPENAI-KEY
AZURE_OPENAI_BASE_ENDPOINT=https://your-resource.openai.azure.com/
AZURE_OPENAI_CHAT_DEPLOYMENT=gpt-4
AZURE_OPENAI_API_VERSION=2024-02-15-preview

# Google
GOOGLE_CLIENT_ID=YOUR-GOOGLE-CLIENT-ID
GOOGLE_CLIENT_SECRET=YOUR-GOOGLE-CLIENT-SECRET

# App Config
APPROVAL_TOKEN=YOUR-RANDOM-APPROVAL-TOKEN
USE MOCK GRAPH=false
FRONTEND_ORIGIN=https://mailmind.yourdomain.com
EOF
```

Replace with your actual values from Section 1:

â€¢ AZURE_CLIENT_ID, AZURE_TENANT_ID, AZURE_CLIENT_SECRET
â€¢ AZURE_OPENAI_API_KEY, AZURE_OPENAI_BASE_ENDPOINT
â€¢ GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET
â€¢ APPROVAL_TOKEN (random hex string)
â€¢ mailmind.yourdomain.com (your actual domain, or use IP for now)

Protect the file:

```
chmod 600 .env
cat .env # Verify it looks correct
```

SECTION 10: Create docker-compose.yml (5 minutes)

In SSH session, in the mailmind directory:

```
cat > docker-compose.yml << 'EOF'
version: '3.8'

services:
  postgres:
    image: postgres:15-alpine
    container_name: mailmind-db
    environment:
      POSTGRES_DB: ${DB_NAME}
      POSTGRES_USER: ${DB_USER}
      POSTGRES_PASSWORD: ${DB_PASSWORD}
    volumes:
      - postgres_data:/var/lib/postgresql/data
    ports:
      - "5432:5432"
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U ${DB_USER}"]
      interval: 10s
      timeout: 5s
      retries: 5
      restart: always
```

NOTE: This is a truncated view. The full docker-compose.yml includes services for backend and nginx. Due to space constraints, see the complete guide in your terminal or reference documentation.

SECTION 13: Start Services (10 minutes)

In SSH session, in the mailmind directory:

```
# Build and start all containers
docker-compose up -d

# Check status (wait 30 seconds for services to start)
sleep 30
docker-compose ps

# You should see:
# mailmind-db postgres:15-alpine Up (healthy)
# mailmind-api mailmind Up (health: starting)
# mailmind-nginx nginx:alpine Up
```

Check backend logs:

```
docker-compose logs backend | head -50

Look for:
INFO: Application startup complete
INFO: Uvicorn running on http://0.0.0.0:8000
```

Test the API:

```
curl -s http://localhost:8000/api/health | head -5
# Should return JSON with "status": "ok"
```

âœ… Services are running!

SECTION 18: Maintenance & Monitoring

View Logs

```
# Real-time backend logs
docker-compose logs -f backend

# Database logs
docker-compose logs -f postgres

# Nginx logs
docker-compose logs -f nginx
```

Update Code

```
# SSH in
ssh -i ssh-key-YYYY-MM-DD.key ubuntu@YOUR-ORACLE-IP

# Pull latest code
cd ~/mailmind
git pull origin main

# Rebuild and restart
docker-compose down
docker-compose up -d --build

# Check it's running
docker-compose ps
```

Database Backup

```
# Backup database
docker-compose exec postgres pg_dump -U mailmind mailmind > ~/mailmind-backup-$(date +%Y%m%d-%H%M%S).sql

# Download backup to local machine
scp -i ssh-key-YYYY-MM-DD.key ubuntu@YOUR-ORACLE-IP:~/mailmind-backup-*.sql ./
```

SECTION 19: Troubleshooting

Backend won't start

```
# Check logs
docker-compose logs backend

# Common issues:
# - DATABASE_URL not set correctly â†’ Check .env file
# - Port 8000 already in use â†’ docker-compose ps
# - Missing Python packages â†’ docker-compose build --no-cache backend

# Rebuild and restart
docker-compose down
docker-compose up -d --build
```

Database won't connect

```
# Check database is running
docker-compose ps

# Check logs
docker-compose logs postgres

# Verify connection from backend logs
docker-compose logs backend | grep -i "database\|connection"
```

Domain not resolving

```
# Check DNS propagation
nslookup api.mailmind.yourdomain.com
dig api.mailmind.yourdomain.com

# If not working, wait 5-10 more minutes or check registrar settings
```

SECTION 20: Final Checklist

- API credentials gathered (Azure, Google)
- Oracle account created
- VM instance running
- Firewall ports open (80, 443, 22)
- SSH access working
- Docker & Docker Compose installed
- Repository cloned
- .env file created with all credentials
- docker-compose.yml created
- Dockerfile created
- nginx.conf created
- Services started with docker-compose up -d
- Backend health check passing
- SSL certificate obtained
- Domain pointing to Oracle IP (or using IP directly)
- Frontend deployed to Vercel
- Triage endpoint tested successfully

☺... Deployment Complete!

Your Live URLs

Frontend: <https://mailmind.yourdomain.com> (or your Vercel URL)

Backend API: <https://api.mailmind.yourdomain.com>

Health Check: <https://api.mailmind.yourdomain.com/api/health>

Database: Internal (postgres:5432)

If you get stuck on any step, refer back to this guide or check the troubleshooting section (SECTION 19).